

Element E: Application of STEM Practices

Project Title: MJ's LAST Stand (Echo Rack)

In our project in order to address the need of a safe and efficient gong stand we will use various STEM subjects in order to ensure our gong stand will be effective and efficient.

Element E - Application of STEM - E1: Application of STEM Concepts to Design:

- 1. Physics:** The fundamental purpose behind the gong stand is to be able to reliably hold a 30-pound gong and additionally withstand a musician repeatedly striking the gong. We have considered the forces of stress that will be inflicted upon the structure of the gong stand, and so we have designed the structure to effectively distribute stress. We have yet to calculate the realistic stresses, but we will in the future.
- 2. Engineering:** The design needs to be able to safely and effectively carry the gong. We will need to apply engineering principles to build the gong effectively and to ensure the safety of the ERHS Band students. Additionally, we will use engineering to ensure the gong has a study base that can withstand and carry its weight. The gong will need to be balanced to ensure it does not slant, potentially endangering the band students. Finally, we will use engineering principles to ensure the gong can be transported easily and safely through the use of wheels.
- 3. Math:** We measured the size of the current gong and applied those measurements to the gong we are making to ensure the gong can fit inside the gong. We will also use math to create a budget for the materials we will use to build our gong stand. Additionally, we need to use math when we are cutting the wood to ensure the wood is cut to certain measurements.

Element E - Application of STEM - E2: Justification of Design Decisions Using STEM Principles:

The construction and design of our musical gong stand was influenced by these STEM principles.

1. Ergonomics/Safety Concerns.

The previous gong stand that was used by the band students generally was unsafe, and since it was carrying a 30-pound gong, it posed a massive safety hazard, since if it would fall, it could seriously injure, and even sprain or break someone's

foot/ankle. We fabricated the design for the new gong stand to be sustainable even after years of use, so even if it is replaced, safety won't be an issue.

2. Cost:

One of the more significant considerations in the project that influenced design was cost. As our research showed, gong stands are usually quite expensive (over hundreds of dollars), so one of our requirements is to maintain a low cost or budget. That means we will be making effective use of relatively low-cost materials and manufacturing methods.

3. Environmentally Friendly/Disposable Materials

The previous invention for the band students, the stand rack, was made out of extremely unstable materials. The wood used was extremely susceptible to mold, and within 2-3 years, it began to rot. Because it becomes a health hazard, as breathing in mold could have detrimental effects, it was extremely hard to dispose of it. Although we are using wood this time, we plan to use things like wood glue to hold materials and generally more environmentally safe/deteriorating materials, and if the worst comes to pass, then it will be easily disposable.

Element E - Application of STEM - E3: Expert Review:

Mr. Szczepanski: As he has previously told me, and what many other professional woodworkers would say, measure twice, cut once. Measurements are a part of math when it comes to our project and STEM. While I can't say he knew too much about the physics part of our project, he did say that pretty much all the wood we'd find at the store would be able to hold the weight of the gong, and we shouldn't worry about it too much. Similarly, without seeing our first 3D models, he would be unsure as to which engineering practices we'd be using. He had no real opinion on the wheels and said that there's no reason that they shouldn't work, especially if we're just buying them from the store. He did tell me, though that supports aren't necessarily needed, but we are free to add them if we're looking for our design to last long.

Element E - Application of STEM - E4: Expert Credentials and Selection:

Mr. Szczepanski: Mr. Szczepanski is a highly trained, expert, self-employed carpenter who runs his own woodworking furniture business. This is his website: [White and Red Works, LLC](#). Although I don't own any of his works, he sometimes spends up to 3 months making luxury furniture items of the highest quality. He knows about wood and wooden products and can help us craft our wooden gong stand. I'm fairly sure he has been woodworking for at least a decade now. Maybe more. He can give us information about what wood we should use and various methods to connect them.

Element E - Application of STEM - E5: Impact of Expert Review on Design Iterations:

Although Mr. Szczepanski has not seen our first iterations of our design (3D models), he did give me some advice beforehand. He claimed that most woods we would buy at a hardware store or pick up from a lumber yard would already be strong enough to hold the weight of the gong and should generally be fine.

Some advice that he gave would be to be careful when buying wood and that you should try to buy wood that's grain is as straight as possible. Also, when assembling the structure, make sure to have the grain edge facing vertically to prevent the wood from sagging over time.

Although Mr. Szczepanski at first wanted us to use specific joints and glue our project together (it is his speciality), once he realized our project was a little bigger than what he had in mind, he recommended using screws to hold everything together and maybe add some support diagonals if we wanted to ensure stability. (Support diagonals are featured in some of our designs.) Now that we have our 3D models, I can try to talk with him this weekend (March 15th and 16th) to get his opinion on the first iteration of our final design.